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SEAT No. _____

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SARDAR PATEL UNIVERSITY
B.Sc.CA & IT/ M. Sc. IT (Integrated) (NC) Examination – 3rd Semester
PS03CIIT02: Advanced Data and File Structures
Thursday, 25th October, 2018

Time: 10:00 am to 1:00 pm

Max. Marks – 70

Q-1 MCQ

[10]

- 1) An array is a _____ data structure.
 - a. Heterogeneous
 - b. Non- Linear
 - c. Unordered
 - d. Homogeneous
- 2) The number of sub tree of a node in a tree is called _____ of a node.
 - a. Degree
 - b. Height
 - c. Level
 - d. Index
- 3) _____ is a set of disjoint trees.
 - a. Path
 - b. Forest
 - c. Directed tree
 - d. Height
- 4) A vertex is _____ if there is no edge connected from any other vertex to the vertex.
 - a. Isolated
 - b. Single
 - c. Self
 - d. Cycle
- 5) A graph is termed as _____ if all the edges in it are labeled with some weights.
 - a. Label graph
 - b. Line graph
 - c. Double graph
 - d. Weighted graph
- 6) A graph G is said to be _____ if each vertex V_i is adjacent to every other V_j in G.
 - a. Multi graph
 - b. Complete graph
 - c. Multiple graph
 - d. Simple graph
- 7) The total numbers of comparisons of selection sort are _____.
 - a. $1/2 * n * (n + 1)$
 - b. $1/2 * n * (n - 1)$
 - c. $2^h - 1$
 - d. $2^{(h - 1)}$
- 8) K-way merging is known as _____.
 - a. Simple merge
 - b. Selection sort
 - c. Multiple merging
 - d. Binary merging
- 9) The _____ is created automatically by the data management routines in the operating system.
 - a. Control area
 - b. Prime area
 - c. Index area
 - d. Overflow area
- 10) _____ indicates how records are distributed over a no. of cylinders.
 - a. Cylinder index
 - b. Prime entry
 - c. Track index
 - d. Master index

(1)

(P.T.O.)

Q-2 Write down the answer of following short questions. (Any Ten)

[20]

- 1) Suppose that each element requires 4 word (byte), the base address of the array a[10] is 650 and lower bound of the array is 0. Then find the address of a[6].
- 2) Define root and leaf of a tree.
- 3) Define : Node and Branch
- 4) What is Sink and source node of a Graph?
- 5) Draw the Binary Tree for $(X-Y) * Z + (P/Q)$.
- 6) Define: strongly and weakly connected Graph.
- 7) List the applications of sorting.
- 8) Write a difference between searching and sorting.
- 9) Define sorting. Also list the sorting techniques.
- 10) What do you mean by file organization? Why it is Required?
- 11) Which criteria are used for selection of file organization?
- 12) Define : Record, Item

Q-3 A) Consider that the base address of the two dimensional array a[5][5] is 2105, each element requires 4 byte (word). Find the address of the element a[3][2] using Row-major order and Column-major order. [05]

B) Explain 2-D array with declaration and initialization. [05]

OR

Q-3 A) List the representation of 2-D array in the memory. Explain any one of them with address calculation formula. [05]

B) Explain Sparse Matrix in detail. [05]

Q-4 A) List various methods of memory representation for Binary tree. Explain any one of them. [05]

B) Draw the binary tree for following expression: [05]

Inorder	F	D	G	B	L	E	A	R	C	S
Postorder	F	G	D	L	E	B	R	S	C	A

OR

Q-4 A) Write an algorithm for Preorder traversal. [05]

B) Draw the binary tree for expression: $(A/B) * (C+D) - (E/F)$. Also find the Inorder traversal of a binary tree. [05]

Q-5 A) Write down the algorithm of Insertion sort. [05]

B) Give a dry run for Selection sort with given : 15, 90, 25, 80, 30 [05]

OR

Q-5 A) Write down the algorithm of Bubble sort. [05]

B) Write down the algorithm of Binary search. [05]

Q-6 A) Explain in detail the structure of Sequential File. [10]

OR

Q-6 A) Explain in detail the structure of Direct File. [10]